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1. Register Complaint (Manual Testing)
2. Run Black Box Tests
3. Run White Box Tests
4. Exit
Select an option: 3
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Control Flow Graph (Register Complaint):
Start
  -> Validate Customer
  -> [Is Customer Valid?]
    -> No -> End (Error)
    -> Yes -> Validate Complaint Description
  -> [Is Description Empty?]
    -> Yes -> End (Error)
    -> No -> Check System Availability
  -> [System Available?]
    -> No -> End (Retry)
    -> Yes -> Assign SLA
  -> Check Agent Availability
  -> [Agent Available?]
    -> Yes -> Assign Agent -> End
    -> No -> Assign Supervisor -> End
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Cyclomatic Complexity = 5
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Independent Paths:
Path 1: Valid Customer -> Valid Description -> System Available -> Agent Available -> End (Assigned)
Path 2: Invalid Customer -> End
Path 3: Empty Description -> End
Path 4: System Unavailable -> End
Path 5: Agent Not Available -> Supervisor -> End
-----
Test Case: TC1
Description: Valid complaint registration
Coverage: Statement Coverage

Input:
  Customer ID: C101
  Complaint Description: Internet not working
  System Available: true
  Agent Available: true

Expected: SUCCESS
Actual: SUCCESS

Result: PASS
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Test Case: TC2
Description: Unknown customer ID
Coverage: Branch Coverage
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Test Case: TC2  
Description: Unknown customer ID  
Coverage: Branch Coverage  
  
Input:  
  Customer ID: C999  
  Complaint Description: Speed issue  
  System Available: true  
  Agent Available: true  
  
Expected: INVALID_CUSTOMER  
Actual: INVALID_CUSTOMER  
  
Result: PASS  
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-----  
Test Case: TC3  
Description: Empty complaint details  
Coverage: Branch Coverage  
  
Input:  
  Customer ID: C101  
  Complaint Description:  
  System Available: true  
  Agent Available: true  
  
Expected: INVALID_COMPLAINT  
Actual: INVALID_COMPLAINT  
  
Result: PASS  
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-----  
Test Case: TC4  
Description: System unavailable  
Coverage: Branch Coverage  
  
Input:  
  Customer ID: C101  
  Complaint Description: Line down  
  System Available: false  
  Agent Available: true  
  
Expected: SYSTEM_UNAVAILABLE  
Actual: SYSTEM_UNAVAILABLE  
  
Result: PASS  
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-----  
Test Case: TC5  
Description: No agent available  
Coverage: Branch Coverage
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Input:

Customer ID: C102
Complaint Description: Installation pending
System Available: true
Agent Available: false

Expected: ESCALATED

Actual: ESCALATED

Result: PASS

Test Case: TC6

Description: Valid complaint different data
Coverage: Condition Coverage

Input:

Customer ID: C102
Complaint Description: Router reboot required
System Available: true
Agent Available: true

Expected: SUCCESS

Actual: SUCCESS

Result: PASS

Test Case: TC7

Description: Blank customer ID
Coverage: Condition Coverage

Input:

Customer ID:
Complaint Description: Billing issue
System Available: true
Agent Available: true

Expected: INVALID_CUSTOMER

Actual: INVALID_CUSTOMER

Result: PASS

Test Case: TC8

Description: Null complaint details
Coverage: Condition Coverage

Input:

Customer ID: C101
Complaint Description: null
System Available: true
Agent Available: true

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Expected: INVALID_COMPLAINT
Actual: INVALID_COMPLAINT
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Result: PASS
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Test Case: TC9
Description: System unavailable with no agent
Coverage: Condition Coverage
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Input:
  Customer ID: C103
  Complaint Description: Service outage
  System Available: false
  Agent Available: false
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Expected: SYSTEM_UNAVAILABLE
Actual: SYSTEM_UNAVAILABLE
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Result: PASS
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Test Case: TC10
Description: Escalation path
Coverage: Path Coverage
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Input:
  Customer ID: C103
  Complaint Description: Account issue
  System Available: true
  Agent Available: false
```

```
Expected: ESCALATED
Actual: ESCALATED
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Result: PASS
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Test Case: TC11
Description: Standard success flow
Coverage: Path Coverage
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Input:
  Customer ID: C101
  Complaint Description: Slow speed
  System Available: true
  Agent Available: true
```

```
Expected: SUCCESS
Actual: SUCCESS
```

```
Result: PASS
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Test Case: TC12  
Description: Additional success flow  
Coverage: Loop Coverage  
  
Input:  
  Customer ID: C102  
  Complaint Description: Signal drops  
  System Available: true  
  Agent Available: true  
  
Expected: SUCCESS  
Actual: SUCCESS  
  
Result: PASS  
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-----  
Test Case: TC13  
Description: Unknown customer alternate  
Coverage: Statement Coverage  
  
Input:  
  Customer ID: C000  
  Complaint Description: Device issue  
  System Available: true  
  Agent Available: true  
  
Expected: INVALID_CUSTOMER  
Actual: INVALID_CUSTOMER  
  
Result: PASS  
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-----  
Test Case: TC14  
Description: Whitespace complaint details  
Coverage: Condition Coverage  
  
Input:  
  Customer ID: C103  
  Complaint Description:  
  System Available: true  
  Agent Available: true  
  
Expected: INVALID_COMPLAINT  
Actual: INVALID_COMPLAINT  
  
Result: PASS  
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-----  
Test Case: TC15  
Description: Success with different data  
Coverage: Statement Coverage  
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Input:
  Customer ID: C103
  Complaint Description: Payment update
  System Available: true
  Agent Available: true

Expected: SUCCESS
Actual: SUCCESS

Result: PASS
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Coverage Types Demonstrated:
✓ Statement Coverage
✓ Branch Coverage
✓ Condition Coverage
✓ Loop Coverage
✓ Path Coverage
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WHITE BOX TESTING SUMMARY
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Total Test Cases: 15
Passed: 15
Failed: 0

Coverage Achieved:

Statement Coverage: 100%
Branch Coverage: 100%
Condition Coverage: 100%
Path Coverage: 100%
Loop Coverage: Complete (0, 1, multiple iterations tested)

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